

# Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

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Why can next-token prediction produce language that appears meaningful? Beginning with tokenization and vector encoding, we study two parts of this question. First, high-dimensional geometry permits exponentially many nearly orthogonal feature directions at a fixed tolerance, although their residual overlaps limit simultaneous readout. Second, polysemy requires us to distinguish classical convex mixtures of word senses, context-dependent occurrence vectors, quantum coherence, and one direction shared by several orthonormal contexts. For ordinary transformers the quantum comparisons are structural and deliberately limited. A final appendix, *A minimal quantum attention model*, presents a finite-dimensional construction with complex amplitudes, learned unitary evolution, optional partial tracing, and phase-sensitive projective Born-rule predictions. Further mathematical details are relegated to undergraduate-level appendices.

## I. WHY DOES A SYNTACTIC MACHINE APPEAR TO UNDERSTAND?

A large language model (LLM) receives strings and returns strings. Yet it can answer questions, continue an argument, translate a passage, and alter its wording when the surrounding text changes. In this behavioral sense it often appears cognizant of what a text is about. How can a machine trained to predict linguistic form display abilities that we ordinarily describe in terms of meaning?

The question recalls John Searle’s Chinese Room thought experiment [1]. A person who does not know Chinese follows formal rules for transforming Chinese symbols and thereby produces answers that appear intelligent to an outside reader. Searle asks whether successful symbol manipulation is sufficient for understanding. We do not try to settle that philosophical question, and fluent behavior will not be used as evidence of consciousness. Our question is narrower and operational: what internal organization allows calculations on tokens to construct and exploit distinctions that human readers recognize as semantic?

There is an important clue. An LLM is not trained on arbitrary strings. Its corpus was produced by people writing about objects, actions, intentions, events, and one another. Regularities of use therefore contain indirect traces of regularities in human experience. The model is never handed a separate object called “meaning.” It is rewarded for predicting how language is used. Successful prediction can favor internal distinctions that track the contexts in which words and constructions occur. Whether this is intrinsic understanding or a powerful simulation remains open; the present chapter concerns the geometry of the representation. It reports no new experiment. The concentration results are established mathematics; their realization in learned LLM feature

dictionaries and the shared-context construction are empirical hypotheses.

Two issues provide a manageable route into that geometry.

First, a representation has a fixed number of independent coordinates, but it may need to distinguish vastly more properties. High-dimensional vector spaces make this geometrically possible because they can contain very many directions that are not exactly orthogonal but have small mutual overlaps. We ask how this *quasi-orthogonality* enlarges expressive capacity and how accumulated overlap—or interference—limits access to the stored distinctions.

Second, one word can have several meanings. The word “bank” may refer to a financial institution, the edge of a river, or a seat. We ask how a single token type and its many occurrences can represent such polysemy. This leads to comparisons with two different quantum ideas: coherent superposition and one direction belonging to several orthonormal bases.

The order of exposition is intentional. Section II begins with tokenization and explains why vector representations are useful. Section III develops the abundance–interference tradeoff in informal terms. Section IV treats polysemy and the quantum comparisons. Appendix D, *A minimal quantum attention model*, presents a quantum model in its own right rather than a reinterpretation of ordinary transformer attention. Additional formal statements and derivations appear in the appendices, where every symbol is defined as it is introduced.

## II. FROM TEXT TO VECTORS

### A. Tokenization: symbols before geometry

The first operation is *tokenization*. A tokenizer divides a character string into units drawn from a finite vocabulary and replaces each unit by an integer identi-

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fier. A token may be a complete word, part of a word, punctuation, a space-bearing fragment, or a byte-level unit. Thus a token is not automatically a word, still less a word’s meaning. Subword schemes assemble rare and previously unseen words from reusable fragments [2].

The assigned integer is merely an address. Identifier 7193 is not semantically closer to identifier 7194 than to identifier 20. The identifiers are therefore used as addresses in a table of vectors on which graded comparison and learned transformation are possible.

### B. A token vector is a starting point, not a complete meaning

A learned lookup table assigns a real vector to every token identifier. Two occurrences of the same identifier begin with the same table row. The model must also know order: “the dog chased the cat” is not equivalent to “the cat chased the dog.” Position is therefore supplied in some form, either by adding positional information or by using another positional mechanism.

The resulting initial vector is not a contextually complete meaning. It has been trained across many occurrences and can reflect what those occurrences share, but later layers construct a new hidden vector for each particular occurrence. The occurrence of “bank” near “loan” can consequently acquire a different hidden state from the same token near “river,” although both began from the same lookup row. This distinction between a token *type* and a contextual token *occurrence* will be essential for polysemy.

Successive transformer layers let each token position combine information from the positions available to it, thereby producing context-dependent vectors. During training, prediction errors adjust persistent model parameters. During ordinary response generation those parameters remain fixed, and each selected token is appended to the text before the next token is predicted.

### C. Why vectors?

Vectors are useful for several reasons. Their coordinates can be adjusted continuously during learning. Matrices can transform them and inner products can compare them. Vector addition permits several contributions to coexist in one state. Most importantly here, lengths, directions, projections, and subspaces provide an economical language for graded features.

Suppose a unit vector  $u$  has been fitted as a direction associated with some property at a chosen layer. Relative to a declared baseline  $h_0$ , the scalar  $u^\top(h - h_0)$  is the signed coordinate of a hidden state  $h$  along that direction. It is the coordinate, not the axis itself, that says on which side of the contrast the state lies and how strongly. In a simplified gender example, the one-dimensional subspace  $\text{span}\{u\}$  supplies a fitted direction

for a gender-associated contrast. The subspace is un-oriented:  $u$  and  $-u$  span the same line. One may orient  $u$  from a feminine-marked sample mean toward a masculine-marked sample mean. Together with  $h_0$ , this defines the affine axis  $h_0 + \text{span}\{u\}$ . Positive and negative projections then describe opposite sides of the same axis; they are not two orthogonal directions. The baseline or a probe intercept defines the zero. Modern contextual representations may use several related directions rather than one universal axis, and the score can change from sentence to sentence [3, 4]. The superscript  $\top$  means transpose, so  $u^\top h$  is the ordinary Euclidean inner product.

This directional interpretation is operational, not metaphysical. It depends on the layer, data, centering, inner product, and extraction method. A probe that reads a property shows that the information is linearly available; it does not by itself prove that the model uses that direction causally. Orthogonality nevertheless has one precise consequence. If two property directions are perpendicular, a component along either direction contributes no cross-talk when the other is read by an inner product. The two directions are therefore *geometrically decoupled for this linear readout*. If one uses the phrase “operational semantic independence,” it means only this absence of readout cross-talk. It does not mean that the corresponding concepts are logically unrelated, statistically independent in the data, or causally independent inside the model.

The reason to emphasize vectors is therefore not that meanings are known in advance to be points in a Euclidean space. It is that a trained vector system can discover useful graded distinctions, combine them, and compare them by simple operations. High dimension adds a striking opportunity: far more approximately separated directions can coexist than there are exactly independent coordinates. That opportunity motivates our first central issue.

## III. MANY DIRECTIONS IN A FIXED-WIDTH SPACE

### A. Exact independence and approximate separation

In a  $d$ -dimensional space, no more than  $d$  nonzero vectors can be mutually orthogonal. This is elementary linear algebra: mutually orthogonal nonzero vectors are linearly independent, and a linearly independent family cannot contain more vectors than the dimension. If semantic distinctions required exactly perpendicular axes, fixed width would impose a severe limit.

Exact orthogonality is stronger than most readouts require. Let  $u$  and  $v$  be unit vectors. Their inner product is the cosine of the angle between them. We call them quasi-orthogonal at tolerance  $\varepsilon$  when  $|u^\top v| \leq \varepsilon$ , where the tolerance is chosen before the comparison. They

are not independent coordinates; they are approximately separated directions.

High-dimensional geometry makes such directions surprisingly abundant. For a uniformly random unit vector in  $d$  real dimensions, the overlap with a fixed unit direction is typically of size  $1/\sqrt{d}$ . Most of the sphere is therefore concentrated near the equator perpendicular to any chosen direction. Lévy’s concentration principle expresses this typicality. The Johnson–Lindenstrauss lemma approaches the same phenomenon from another side: a finite collection of points can be placed in a dimension proportional to the logarithm of the number of points while approximately preserving all pairwise distances. Read in the constructive direction, these results show that at any fixed nonzero angular tolerance a  $d$ -dimensional representation space admits at least exponentially many quasi-orthogonal directions [5–7].

The word *exponential* is important, but so are its qualifications. It counts directions available at the chosen tolerance, not a second vector-space dimension. It is a geometric possibility, not proof that a trained model uses all those directions. The allowed overlap must be stated, and an exponentially large dictionary is not automatically readable when many of its elements are active at once. Appendix A introduces the notation, and Appendix B gives the concentration, union-bound, and Johnson–Lindenstrauss calculations.

### B. The price of abundance: interference

The simplest calculation already shows the access problem. Suppose a hidden state contains two unit feature directions,

$$h = au + bv. \quad (1)$$

Reading the first feature by an inner product gives  $u^\top h = a + b(u^\top v)$ . The desired coefficient is  $a$ ; the second term is cross-talk from the other feature. Exact orthogonality removes it, while quasi-orthogonality bounds its size by  $\varepsilon|b|$ .

Many small unwanted terms can nevertheless accumulate. If  $k$  comparably sized, random-like features are active and their signs do not align systematically, their mean squared errors add as in the Pythagorean theorem. The resulting root-mean-square (RMS) interference is of order  $\sqrt{k/d}$  relative to one unit-scale target. This is an average random-code benchmark, not a guarantee for every learned dictionary. In the worst case, aligned cross-terms can add rather than cancel.

The useful operating region is consequently task dependent. There should be enough distinguishable directions for the features a task needs, but not so many simultaneously active, strongly overlapping directions that the chosen readout fails. There is no universal numerical boundary determined by  $d$  alone. It also depends on which features coactivate, their strengths, the decoder, and the tolerated error. A learned model may use its

angular budget selectively, placing greater separation between features that frequently need to be read together.

### C. A limited analogy with geometric decoherence

Small overlaps also occur in quantum theory. When alternatives of a quantum system become correlated with nearly orthogonal environmental states, the off-diagonal terms of the reduced state are multiplied by the corresponding environmental overlaps. High dimension can make those factors very small when the conditional states or their relative dynamics satisfy an explicit typicality assumption; dimension alone is not enough, since the conditional states could remain identical. The author’s companion analysis studies this purely geometric contribution to decoherence [8].

The LLM analogy is narrow. Quasi-orthogonal semantic directions suppress linear cross-talk in much the way that nearly orthogonal environmental branch states suppress observable coherence terms. We call this *geometric decoherence*: unwanted contributions become small because the relevant inner products are small. It is not physical quantum decoherence. An LLM has no required system–environment split, unitary branch dynamics, partial trace, or local loss of access to phase relations after environmental degrees of freedom are traced out. High dimension supplies shared geometry, not a shared physical mechanism [9].

The word *orthogonal* also has different operational meanings here. For the LLM readout it means geometric decoupling: absence of linear cross-talk in the chosen inner product. Within one quantum measurement context, orthogonal rays represent mutually exclusive outcomes. The common inner-product geometry does not make these interpretations identical.

This qualified comparison is still useful. In both settings, exact orthogonality is an ideal limit; approximate orthogonality can already make alternatives operationally difficult to confuse; and the number of pairwise small overlaps is not the same as the number of alternatives that can be reliably recovered together.

## IV. POLYSEMY: ONE FORM, SEVERAL MEANINGS

### A. A sense is not a single feature

The simplest vector account of polysemy is entirely classical. A static word vector may be modeled as a real, frequency-weighted convex combination of vectors associated with its senses. A contextual model goes further by constructing a different real occurrence vector from the available text. Neither construction is, without additional operational structure, a coherent quantum superposition.

A *sense* is a contextually appropriate meaning of a lexical item. The financial, river-edge, and seat meanings of “bank” are different senses. The banking or tilting of an aircraft supplies another. A semantic feature is narrower: it may mark animacy, tense, number, sentiment, or one fitted gender contrast. Several features can contribute to one sense, and one feature can recur in many words and senses. Polysemy is therefore not solved merely by finding one direction for every dictionary definition.

It is also important to say which representation is being discussed. A static embedding assigns one stored vector to a word type. A transformer has one initial lookup row for a token type but constructs different hidden vectors for its occurrences. These are related but not identical problems.

### B. A static vector as a corpus-level mixture

In a static system such as word2vec, the same stored type vector is used in every occurrence. This may appear poorly suited to polysemy. Under a particular generative model, however, Arora and collaborators establish an approximate relation between that one type vector and hypothetical sense vectors [10]. The following three-sense equation illustrates the corresponding finite-mixture idea:

$$\begin{aligned} v_{\text{bank}} &\approx p_{\text{fin}} v_{\text{fin}} + p_{\text{river}} v_{\text{river}} + p_{\text{seat}} v_{\text{seat}}, \\ p_{\text{fin}}, p_{\text{river}}, p_{\text{seat}} &\geq 0, \\ p_{\text{fin}} + p_{\text{river}} + p_{\text{seat}} &= 1. \end{aligned} \quad (2)$$

The nonnegative coefficients summarize how frequently the senses occur in the corpus. The equation is an approximate classical convex combination and a population-level summary. It does not decide what one occurrence means, nor does one observed sum uniquely determine its hidden summands.

Arora et al. also expressed each word vector using only a few reusable, topic-like directions, which they called discourse atoms. This sparse description can help recover recurring components across many words. It does not show that all sense vectors or atoms are mutually quasi-orthogonal, and the atoms need not correspond one-to-one with dictionary senses. The geometric results of Sec. III establish what is possible at a chosen tolerance; they do not establish what a learned lexical dictionary actually contains.

### C. Contextual occurrences

A static mixture summarizes a word type across a corpus; it does not determine the meaning of one occurrence. Modern transformers begin occurrences of the same token type from the same lookup vector, but the available text changes the vectors constructed later. Thus “bank” in a financial passage and “bank” in a river passage can

acquire different occurrence vectors. In a causal text generator, a position can use only earlier text; a clarifying word that appears later can influence later positions, but it cannot retrospectively change the earlier state.

For intuition, one may imagine the context as changing the relative importance of several sense contributions. This need not be the transformer’s literal internal algorithm, and an ambiguous sentence or pun may preserve more than one possibility. The cautious statement is therefore that context *transforms* an occurrence representation. It does not perform a quantum measurement or collapse a physical superposition. Appendix C gives a compact mathematical version of this distinction.

### D. Why this is not automatically quantum coherence

Quantum theory uses the word *superposition* in a stronger operational sense. A pure state combines alternatives with probability amplitudes that may be real or complex. They need not be nonnegative and need not sum to one; instead, the sum of their squared magnitudes is one. A pure-state superposition is therefore not a convex average. Relative phase can affect later measurement probabilities when alternatives are recombined in another basis, and the Born rule turns amplitudes into probabilities. A classical frequency average such as Eq. (2) provides none of this. Its coefficients are frequencies, not amplitudes; no phase-sensitive transformation or interference experiment is specified; and its sense vectors are not declared orthonormal measurement outcomes.

Why not impose a full quantum-style superposition? One may certainly propose a new mathematical model that does so. But the proposal would have to state the sense basis, amplitudes, relative phases, allowed transformations, measurement rule, and an experiment in which phase changes alter observable predictions. Without those ingredients, replacing the classical weights by symbols resembling quantum amplitudes merely renames a vector sum. Ordinary transformer processing supplies no Born rule and no physical collapse. The explicit pure-state and density-matrix comparison is given in Appendix C. Appendix D, *A minimal quantum attention model*, supplies such a specification, including a phase-sensitive “bank” experiment. It is not a reinterpretation of ordinary transformer attention.

### E. A different quantum analogy: shared contexts

Quantum logic suggests a comparison that does not rely on vector addition. An orthonormal measurement context can be represented by mutually orthogonal rays, or equivalently by rank-one projectors that sum to the identity. In dimension at least three, distinct contexts may contain the same projector while differing in their

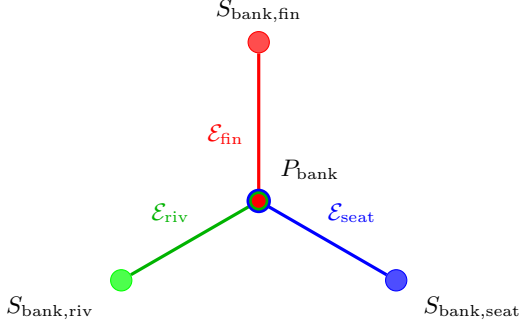


FIG. 1. Structural incidence schematic for “bank.” The central projector  $P_{\text{bank}}$  is shared. Each colored arm shows only a fragment of one context hyperedge; unshown orthogonal projectors complete the corresponding basis. An outer node  $S_{\text{bank},s}$  summarizes a residual semantic sector, not necessarily one rank-one outcome. The drawing depicts shared incidence, not the vector sum in Eq. (2).

other members. This is an incidence relation: one element belongs to several contexts.

The diagram below borrows that pattern for a speculative representation of “bank.” Its center is a fixed lexical projector. Each colored arm belongs to a different larger context, whose remaining directions carry context-specific information. The same lexical element is retained, while the financial, river-edge, and seat contexts differ around it.

This analogy has a sharp limit. The shared projector alone cannot say which sense is intended. If a quantum system were prepared exactly in that shared ray, a measurement in any basis containing it would return the shared outcome with certainty. More generally, for a fixed quantum state the Born probability of the same projector does not depend on which compatible projectors complete the basis. Linguistic interpretation is different in the relevant respect: it is supposed to depend on the available text.

The diagram can therefore motivate only an optional classical hypothesis: a contextual occurrence vector might retain a stable nonzero lexical component while placing sense-dependent information in accompanying directions. Such a decomposition is mathematically easy to construct; its scientific content would lie in stability, predictive usefulness, and improvement over simpler untied contextual models. The analogy is structural, not a claim that polysemy is a quantum measurement. Appendix C gives the shared-projector construction and explains why quantum probability rules do not transfer to linguistic context.

If the proposed lexical anchor is unstable or provides no advantage over an equally flexible model without it, this optional shared-context hypothesis fails while the ordinary contextual account of polysemy remains intact.

## V. CONCLUSION

Searle’s Chinese Room asks whether correct manipulation of symbols amounts to understanding. We have not answered that philosophical question. We have instead followed one operational path from tokens to meaning-sensitive behavior.

Token identifiers first become vectors because vectors support graded coordinates, learned transformations, comparison, and combination. Their high-dimensional geometry addresses the first problem considered here. A fixed-width representation space can accommodate exponentially many quasi-orthogonal directions at a chosen tolerance, even though it has only  $d$  exactly independent coordinates. This is expressive room, not cost-free access. Residual overlaps create interference, especially when many relevant features are active together.

The resulting overlap suppression has a limited quantum counterpart. Small inner products suppress cross-terms both in linear semantic readout and in the environmental overlaps that enter quantum decoherence. The shared geometry justifies the phrase geometric decoherence, while the absence of quantum dynamics, relative phases, and environmental tracing in ordinary transformer processing prevents a physical identification.

Polysemy exposes different limits of the quantum comparison. A static word vector may approximate a frequency-weighted convex mixture of sense vectors, and a transformer constructs context-dependent occurrence vectors. Neither is thereby a coherent quantum superposition, and contextualization is not quantum collapse. The star diagram retains only a quantum-logical incidence idea: one lexical projector may be shared by several orthonormal contexts. That optional constraint must earn its place empirically; the common ray alone does not select a meaning.

At present, the relation between ordinary transformer LLMs and quantum mechanics is best regarded as a limited analogy of inner-product geometry rather than a common physical or semantic theory. Language-model activations inhabit finite-dimensional real inner-product spaces; standard quantum mechanics represents pure states as rays in a complex Hilbert space and adds observables, tensor composition, and the Born rule. Even shared terms have different meanings: linguistic context is surrounding text, whereas a quantum context is a family of compatible observables; “interference” here means linear readout cross-talk, whereas quantum interference is phase-sensitive.

Quasi-orthogonality is nevertheless a genuine common geometric motif. In a language model it offers a possible capacity mechanism for many approximately distinguishable features, subject to cross-talk and empirical confirmation. In quantum theory, near-orthogonality of conditional environmental states can, under appropriate dynamics, suppress reduced-state coherences and thereby contribute to quasi-classical behavior. It does not by itself select a pointer basis, produce definite outcomes, or

establish stable classical records. The paper thus identifies geometric resources for semantic discrimination, not a complete theory of machine understanding.

### Appendix A: Formal companion to *From text to vectors*

This appendix gives a minimal mathematical version of Sec. II. It distinguishes strings, token identifiers, and vectors before discussing semantic directions.

#### 1. Tokenization and lookup

Let  $\mathcal{S}$  be the set of finite character strings. Let  $\mathcal{V} = \{1, \dots, V\}$  be a vocabulary containing  $V$  token identifiers. Each identifier is an integer label. A tokenizer is a fixed map

$$\text{Tok} : \mathcal{S} \longrightarrow \bigcup_{T \geq 0} \mathcal{V}^T. \quad (\text{A1})$$

If  $\text{Tok}(\sigma) = (t_1, \dots, t_T)$ , then the input string  $\sigma$  has become a sequence of  $T$  identifiers. The union in Eq. (A1) allows strings to produce sequences of different lengths. A detokenizer reverses the assembly, subject to the system's stated normalization rules.

Let  $d_{\text{model}}$  be the architectural width of the model's hidden stream. The notation  $\mathbb{R}^m$  means the set of real column vectors with  $m$  coordinates. An embedding table is a real matrix  $E \in \mathbb{R}^{V \times d_{\text{model}}}$ . Its row  $E[t_i]$  is the initial lookup vector for identifier  $t_i$ . In a simple additive positional scheme,

$$h_i^{(0)} = E[t_i] + p_i, \quad h_i^{(0)}, p_i \in \mathbb{R}^{d_{\text{model}}}. \quad (\text{A2})$$

Here  $p_i$  supplies the position and the superscript (0) means the stage before the first transformer layer. Some architectures incorporate position inside later comparison operations rather than literally adding  $p_i$ ; the conceptual role is the same. The identifier  $t_i$ , the table row  $E[t_i]$ , and the contextual state produced at a later layer are three different objects.

The geometric analysis need not use all  $d_{\text{model}}$  architectural coordinates. A declared preprocessing map may center the observed states and remove nearly unused directions. It may also rescale the retained principal directions to a declared common variance, a step called whitening. We use  $d \leq d_{\text{model}}$  for the rank of the resulting representation space. Any numerical claim involving  $d$  must therefore report the model, layer, sample, centering, rank threshold, and inner product used to obtain it.

#### 2. Projection onto a fitted feature direction

Let  $u \in \mathbb{R}^d$  be a unit vector, so  $u^\top u = 1$ . The orthogonal projection of a state  $h$  onto the line through  $u$

is

$$P_u h = (u^\top h)u, \quad P_u = uu^\top. \quad (\text{A3})$$

The scalar  $u^\top h$  is its signed coordinate. The matrix  $P_u$  is called a projector because direct multiplication gives  $P_u^2 = P_u$ . The remainder  $h - P_u h$  is perpendicular to  $u$ , since  $u^\top (h - P_u h) = 0$ . This is the elementary orthogonal projection theorem in finite-dimensional form.

An oriented contrast needs a baseline. Suppose two training samples have means  $\mu_A$  and  $\mu_B$  in the same representation space and  $\mu_A \neq \mu_B$ . Define

$$g = \frac{\mu_A - \mu_B}{\|\mu_A - \mu_B\|}, \quad h_0 = \frac{\mu_A + \mu_B}{2}, \quad (\text{A4})$$

$$\alpha_g(h) = g^\top (h - h_0).$$

The unit vector  $g$  orients the contrast,  $h_0$  declares its midpoint, and  $\alpha_g(h)$  is the signed coordinate. Substitution shows

$$\alpha_g(\mu_A) = \frac{\|\mu_A - \mu_B\|}{2}, \quad \alpha_g(\mu_B) = -\frac{\|\mu_A - \mu_B\|}{2}. \quad (\text{A5})$$

Swapping the two groups reverses  $g$  and every sign but does not change the unoriented line. A score near zero means that the coordinate along  $g$  is near the midpoint; it does not imply that the full vector is near  $h_0$ , because a large perpendicular component may remain.

The same construction can illustrate a fitted gender-associated contrast, but it does not define gender and need not capture all gender-related information. A linear classifier provides a related alternative: if its decision boundary is  $w^\top h + \beta_0 = 0$  with  $w \neq 0$ , then  $(w^\top h + \beta_0)/\|w\|$  is signed Euclidean distance from that hyperplane. Either construction must be fitted on training examples and evaluated on held-out examples.

### Appendix B: Formal companion to the geometric argument

The main text used three counts. The integer  $d$  is the exact dimension of the selected vector space. The integer  $M$  is the number of feature or code directions in a declared dictionary. The integer  $k$  is the number of those directions appreciably active in one state. Only  $d$  is a vector-space dimension.

#### 1. Exact orthogonality and quasi-orthogonality

Let  $w_1, \dots, w_M \in \mathbb{R}^d$  be unit vectors. The family is  $\varepsilon$ -quasi-orthogonal when

$$|w_i^\top w_j| \leq \varepsilon \quad \text{whenever } i \neq j, \quad 0 < \varepsilon < 1. \quad (\text{B1})$$

The absolute value treats a direction and its negative as the same unoriented axis for purposes of overlap.

When  $\varepsilon = 0$ , the vectors are mutually orthogonal. To see why there can then be at most  $d$ , suppose  $a_1 w_1 + \dots + a_M w_M = 0$ . Taking the inner product with  $w_j$  leaves  $a_j = 0$ , because every cross-term vanishes and  $w_j^\top w_j = 1$ . All coefficients are zero, so the vectors are linearly independent. The definition of dimension now gives  $M \leq d$ .

For  $\varepsilon > 0$ , the vectors need not be linearly independent. The largest possible  $M$  becomes a finite spherical-code problem at the chosen tolerance. Its cardinality is sometimes informally called quasi-dimensionality, but no additional linear dimensions have appeared.

## 2. Why typical overlaps shrink with dimension

For a random quantity  $Y$ ,  $\mathbb{E}[Y]$  denotes its average. For an event  $A$ ,  $\mathbb{P}(A)$  denotes its probability. A unit vector chosen uniformly from the sphere has no preferred direction: rotating the whole experiment does not change its probability law.

Fix a unit vector  $u$  and choose  $X$  uniformly from the unit sphere in  $\mathbb{R}^d$ . Rotational symmetry lets us take  $u$  to be the first coordinate direction. The overlap  $u^\top X$  is then  $X_1$ . Symmetry makes all coordinate squares have the same expectation. Since  $X_1^2 + \dots + X_d^2 = 1$ , taking expectations gives

$$1 = d \mathbb{E}[X_1^2], \quad \mathbb{E}[(u^\top X)^2] = \frac{1}{d}. \quad (\text{B2})$$

The RMS overlap is the square root of the mean squared overlap, hence  $1/\sqrt{d}$ . This elementary calculation gives a scale, not a tail probability and not a guarantee for learned vectors.

Lévy's concentration lemma supplies the tail. One convenient consequence is that universal positive constants  $c$  and  $C$  exist such that

$$\mathbb{P}\{|u^\top X| > \varepsilon\} \leq C \exp(-cd\varepsilon^2) \quad (\text{B3})$$

for a fixed unit  $u$  and a relevant fixed tolerance. The function  $x \mapsto u^\top x$  is 1-Lipschitz because the Cauchy–Schwarz inequality gives  $|u^\top(x - y)| \leq \|x - y\|$ . Lévy's lemma says that a Lipschitz function on a high-dimensional sphere is unlikely to deviate far from its typical value. Here 1-Lipschitz means that changing the input by a distance  $r$  changes the function value by at most  $r$  [5, 11].

## 3. From one pair to an exponential family

Choose  $M$  directions independently and uniformly. There are  $M(M-1)/2$  unordered pairs. The union bound states that the probability of at least one bad pair is no larger than the sum of the bad-pair probabilities. Combining it with Eq. (B3) gives

$$\mathbb{P}\left\{\max_{i \neq j} |w_i^\top w_j| > \varepsilon\right\} \leq \frac{CM(M-1)}{2} \exp(-cd\varepsilon^2). \quad (\text{B4})$$

Choose the greatest integer  $M$  not exceeding  $\exp(c'\varepsilon^2 d)$ , where  $c'$  is sufficiently small and  $d$  is sufficiently large. Then the right side is below one. Therefore at least one family of that exponential size has every overlap within tolerance. This is an existence lower bound and a random-code benchmark, not a universal maximum and not proof that an LLM learns such a code.

Solving the exponent in Eq. (B4) for the overlap tolerance shows that the union-bound crossover is a constant multiple of  $\sqrt{\log M/d}$  when  $\log M$  is small compared with  $d$ . The union bound supplies a high-probability upper scale, not by itself a matching lower law. This remains a null-model estimate rather than a hard capacity wall.

## 4. The Johnson–Lindenstrauss route

The Johnson–Lindenstrauss lemma states that a set of  $M+1$  points can be mapped into  $\mathbb{R}^d$  while preserving all squared distances within relative error  $\delta$  when

$$d \geq C_{\text{JL}} \delta^{-2} \log(M+1), \quad 0 < \delta < 1, \quad (\text{B5})$$

where  $C_{\text{JL}}$  is a numerical constant and  $\log$  is the natural logarithm [6, 7]. This is a theorem about a prescribed finite set, not random semantic features.

To obtain a quasi-orthogonal code, take the origin and  $M$  mutually orthogonal standard coordinate vectors  $e_i$  in a sufficiently large starting space. Let  $\Phi$  be a linear Johnson–Lindenstrauss map and define  $y_i = \Phi e_i$ . Including the origin controls the image lengths. The polarization identity

$$2y_i^\top y_j = \|y_i\|^2 + \|y_j\|^2 - \|y_i - y_j\|^2 \quad (\text{B6})$$

turns preservation of lengths and distances into preservation of inner products. Before normalization their pairwise absolute inner products are at most  $2\delta$ ; because every squared length is at least  $1 - \delta$ , the normalized overlaps are at most  $2\delta/(1 - \delta)$ . For a fixed target dimension  $d$ , choose an integer  $M$  satisfying  $M+1 \leq \exp(c\delta^2 d)$ , with  $c$  small enough that Eq. (B5) holds. This constructs an exponential-size family.

Lévy concentration, random spherical coding, and the Johnson–Lindenstrauss lemma are thus related views of one high-dimensional phenomenon. Their exponential factors must not be multiplied as though they were independent mechanisms.

## 5. A deterministic overlap floor

Random coding gives a typical construction. The Welch bound gives a deterministic limitation. Put the vectors into the columns of a matrix  $W \in \mathbb{R}^{d \times M}$  and form the Gram matrix  $G = W^\top W$ . Its entry  $G_{ij} = w_i^\top w_j$  records every pairwise overlap. The diagonal entries are

one, so the trace of  $G$  is  $M$ . The matrix has rank at most  $d$  and nonnegative eigenvalues.

Let  $\lambda_1, \dots, \lambda_r$  be its nonzero eigenvalues, where  $r \leq d$ . Cauchy–Schwarz gives

$$M^2 = \left( \sum_{\ell=1}^r \lambda_\ell \right)^2 \leq r \sum_{\ell=1}^r \lambda_\ell^2 \leq d \sum_{\ell=1}^r \lambda_\ell^2. \quad (\text{B7})$$

The sum of squared matrix entries equals the sum of squared eigenvalues for a real symmetric matrix. Removing the  $M$  diagonal ones and averaging the remaining entries yields

$$\max_{i \neq j} |w_i^\top w_j|^2 \geq \frac{M-d}{d(M-1)} \quad (M > d). \quad (\text{B8})$$

This is the Welch bound [12]. It says that once the dictionary is overcomplete, some nonzero overlap is unavoidable. It does not say which pair overlaps or how damaging that pair is for a particular task.

## 6. Two-feature and many-feature readout

Let a hidden state be represented by

$$h = \sum_{i \in A} a_i w_i + e. \quad (\text{B9})$$

Here  $A$  is the active set,  $a_i$  are scalar feature coefficients, and  $e$  contains components omitted by this chosen dictionary. Reading feature  $j \in A$  by correlation gives

$$w_j^\top h = a_j + \sum_{\substack{i \in A \\ i \neq j}} a_i (w_j^\top w_i) + w_j^\top e. \quad (\text{B10})$$

The first term is desired; the sum contains geometric cross-talk. Equation (1) is the case  $A = \{1, 2\}$  and  $e = 0$ .

If the directions are  $\varepsilon$ -quasi-orthogonal, the triangle inequality gives the worst-case bound

$$\left| \sum_{\substack{i \in A \\ i \neq j}} a_i (w_j^\top w_i) \right| \leq \varepsilon \sum_{\substack{i \in A \\ i \neq j}} |a_i|. \quad (\text{B11})$$

This bound allows every unwanted term to have the same sign. Under a different, explicitly random-like model, assume instead that the cross-terms are centered, have zero cross-moments, and satisfy  $\mathbb{E}[(w_j^\top w_i)^2 \mid w_j] = 1/d$ . Treat the active set and coefficients as fixed and condition on  $w_j$ . Variances then add:

$$\mathbb{E} \left[ \left( \sum_{\substack{i \in A \\ i \neq j}} a_i w_j^\top w_i \right)^2 \mid A, (a_i), w_j \right] = \frac{1}{d} \sum_{\substack{i \in A \\ i \neq j}} a_i^2. \quad (\text{B12})$$

For  $k-1$  interfering features of comparable unit coefficient, the square root is  $\sqrt{(k-1)/d}$ . This is the RMS

law quoted in the main text. Centering or zero cross-moments is essential; an informal claim of uncorrelated contributions would be insufficient without their means being defined.

The contrast between Eqs. (B11) and (B12) explains why no universal Goldilocks boundary follows from  $d$  and  $M$  alone. A task must also specify active coefficients, coactivation frequencies, a decoder, and an accepted error. Overlap between two features that never need to be recovered together may have little cost.

## 7. The common algebra behind geometric decoherence

We first introduce the notation used in this comparison. A pure quantum state may be represented by a normalized complex column vector; vectors that differ only by an overall phase represent the same physical ray. Dirac notation writes such a vector as a *ket*  $|v\rangle$  and its conjugate transpose as the *bra*  $\langle v|$ . Thus  $\langle v|w\rangle$  is an inner product, whereas  $|v\rangle\langle w|$  is a matrix called an outer product. A density matrix is a positive semidefinite matrix of trace one; the trace is the sum of its diagonal entries. The partial trace over an environment is the linear operation that discards the environmental coordinates while preserving all expectation values of measurements made only on the system.

For comparison with physical decoherence, let  $\{|s_i\rangle\}$  be orthonormal alternatives of a quantum system. Suppose a system–environment interaction produces the pure joint state

$$|\Psi\rangle = \sum_i c_i |s_i\rangle |E_i\rangle, \quad \sum_i |c_i|^2 = 1. \quad (\text{B13})$$

The normalized  $|E_i\rangle$  are conditional environmental states. Observing only the system means taking a partial trace over the environment. The reduced system state is

$$\rho_S = \sum_{i,j} c_i c_j^* \langle E_j | E_i \rangle |s_i\rangle \langle s_j|. \quad (\text{B14})$$

Thus every off-diagonal term with  $i \neq j$  is multiplied by an environmental overlap. When the relevant  $|E_i\rangle$  become nearly orthogonal, those coherence terms become small for system-only observations. This is the standard algebraic core of environmental decoherence [9].

Near-orthogonality of the  $|E_i\rangle$  requires a stated assumption about the interaction or branch-state ensemble; high dimension alone does not produce it. Moreover, many small off-diagonal terms can accumulate. Therefore, bounding each pair separately does not by itself show that the whole reduced matrix is close to the matrix obtained by deleting all off-diagonal terms.

Compare Eq. (B14) with Eq. (B10). The quantum coherence term is suppressed by  $\langle E_j | E_i \rangle$ ; the classical readout error is suppressed by  $w_j^\top w_i$ . Both are small-overlap effects. Only the former follows from a joint quantum

state and a partial trace. The latter is ordinary cross-talk in a real linear readout. This is why geometric decoherence is an analogy in the LLM setting rather than a claim of physical decoherence.

## 8. Conditional tensor-product amplification

One further scaling is mathematically correct but not implied for an ordinary transformer. If a separate model genuinely has  $n$  tensor factors, each of fixed local dimension  $q \geq 2$ , then its exact Hilbert-space dimension is

$$d = q^n. \quad (\text{B15})$$

Substitution into the exponential code bound gives

$$M \geq \exp(c\varepsilon^2 q^n) - 1, \quad (\text{B16})$$

which is doubly exponential in the number of factors  $n$  at fixed  $q$  and  $\varepsilon$ , for sufficiently large  $n$ . This is a code cardinality, not an additional Hilbert-space dimension. A real code is also a valid subset of the corresponding complex representation space. Standard LLM residual width is not known to equal an accessible  $q^n$  tensor-product representation space, so Eq. (B16) is not an ordinary LLM capacity claim.

## Appendix C: Formal companion to the polysemy comparison

### 1. Static mixture and contextual reweighting

Let  $t$  denote a word type, or a token that corresponds to a complete lexical item, and let  $s \in \{1, \dots, S_t\}$  index its candidate senses. Let  $v_{t,n}^{\text{type}}$  be the static embedding estimated from an amount of corpus data indexed by  $n$ . Under the generative assumptions used by Arora et al., this estimate converges in the corpus limit to a population vector  $\bar{v}_t$ . That limit is approximately related to sense-specific embeddings  $q_{t,s}$  by

$$\begin{aligned} v_{t,n}^{\text{type}} &\longrightarrow \bar{v}_t & (n \rightarrow \infty), \\ \bar{v}_t &\approx \sum_{s=1}^{S_t} \pi_t(s) q_{t,s}, & \sum_s \pi_t(s) = 1. \end{aligned} \quad (\text{C1})$$

If  $f_{t,s}$  is the corpus count of sense  $s$  among occurrences of type  $t$ , then  $\pi_t(s) = f_{t,s} / \sum_r f_{t,r}$  is its relative frequency. The first relation is the asymptotic convergence statement; the second is the model-dependent approximation to a frequency-weighted sense sum. Arora et al. prove the displayed relation explicitly for two senses; the finite multisense expression is their general linearity assertion and follows from the same conditioning argument [10]. Keeping the two relations separate avoids attributing every modeling residual to finite-sample error.

An interpretable contextual extension replaces the fixed frequency prior by weights inferred from the available text  $x$ :

$$\begin{aligned} q_t^{\text{mix}}(x) &= \sum_{s=1}^{S_t} p_\theta(s | x, t) q_{t,s}, \\ p_\theta(s | x, t) &\geq 0, & \sum_s p_\theta(s | x, t) &= 1. \end{aligned} \quad (\text{C2})$$

The subscript  $\theta$  denotes fitted parameters of the contextual weighting rule. Equation (C2) is a useful classical baseline. An actual transformer guarantees only that some learned map  $F_\ell$  produces an occurrence state

$$h_\ell(t | x) = F_\ell(E[t], x) \quad (\text{C3})$$

at layer  $\ell$ . It need not implement the explicit convex mixture in Eq. (C2).

### 2. Quantum pure superposition and incoherent mixture

Using the Dirac notation introduced in Appendix B 7, let  $\{|s\rangle\}$  now denote orthonormal quantum alternatives. A normalized pure state is

$$|\psi\rangle = \sum_s c_s |s\rangle, \quad \sum_s |c_s|^2 = 1. \quad (\text{C4})$$

The coefficients  $c_s$  are amplitudes. They may be real or complex; quantum coherence does not require every amplitude to be non-real. What matters is that relative signs or phases can affect probabilities after a change of measurement basis.

The density matrix of the pure state is

$$|\psi\rangle\langle\psi| = \sum_{s,r} c_s c_r^* |s\rangle\langle r|. \quad (\text{C5})$$

The star denotes complex conjugation. Terms with  $s \neq r$  are off-diagonal coherences in this basis. By contrast, an incoherent quantum state formed by classical probabilistic mixing of the same orthogonal alternatives is represented by

$$\rho_{\text{mix}} = \sum_s p_s |s\rangle\langle s|, \quad p_s \geq 0, \quad \sum_s p_s = 1. \quad (\text{C6})$$

This positive semidefinite, trace-one matrix has no off-diagonal terms in the displayed basis. Under its intended lexical interpretation, the Arora sum in Eq. (C1) is neither Eq. (C4) nor Eq. (C6): it is a real vector average in a lexical embedding space. Ordinary Euclidean cross-terms do not establish quantum interference.

### 3. The optional shared-projector construction

Let  $r_t \in \mathbb{R}^d$  be a unit lexical anchor and let

$$P_t = r_t r_t^\top \quad (\text{C7})$$

be its rank-one projector. For linguistic context class  $s$ , choose  $m_s$  orthonormal residual directions perpendicular to  $r_t$  and place them in the columns of

$$B_{t,s} = [u_{t,s,1} \cdots u_{t,s,m_s}], \quad (C8)$$

$$B_{t,s}^\top B_{t,s} = I_{m_s}, \quad B_{t,s}^\top r_t = 0.$$

The identity matrix  $I_{m_s}$  has  $m_s$  rows and columns. The columns of  $B_{t,s}$  form an orthonormal partial frame. The Gram–Schmidt or orthonormal-extension theorem permits it to be completed to a full basis of  $\mathbb{R}^d$ . Put  $P_{t,s,j+1} = u_{t,s,j} u_{t,s,j}^\top$  for the displayed residual directions and use the additional basis directions for the remaining projectors. In projector language, a completed context has the form

$$\mathcal{E}_{t,s} = \{P_t, P_{t,s,2}, \dots, P_{t,s,d}\}, \quad P_t + \sum_{j=2}^d P_{t,s,j} = I_d. \quad (C9)$$

The  $P_{t,s,j}$  are mutually orthogonal rank-one projectors. Different hyperedges  $\mathcal{E}_{t,s}$  may share  $P_t$  and differ in their other projectors. The orthonormal completion is nonunique, so its mere existence is automatic mathematics rather than an empirical discovery.

A strict shared-anchor hypothesis for contextual states could be written

$$h_\ell(t | x) = a_t r_t + B_{t,s(x)} z_{t,s(x)}(x) + e_t(x), \quad (C10)$$

$$P_t e_t(x) = 0, \quad B_{t,s(x)}^\top e_t(x) = 0.$$

Here  $s(x)$  is a context class inferred from text,  $z_{t,s(x)}(x)$  contains context-dependent residual coordinates, and  $a_t \neq 0$  is required to be the same across contexts. The two displayed orthogonality conditions put  $e_t(x)$  outside both the anchor and the selected residual sector, so components cannot be shifted arbitrarily between  $B_{t,s(x)} z_{t,s(x)}(x)$  and  $e_t(x)$ . Only the component  $a_t r_t$  is fixed; the complete occurrence vector generally changes. If the entire vector were fixed, the context would have to be stored elsewhere and that vector alone would not encode polysemy.

For a fixed quantum state  $\rho$ , the Born probability assigned to the shared projector is

$$p(P_t | \rho) = \text{tr}(\rho P_t). \quad (C11)$$

No term in this expression names the other projectors that complete the measurement context. Gleason’s theorem, under its additivity hypotheses in real or complex dimension at least three, derives this density-operator form for projector probabilities [13]. This limited probability consistency does not say that quantum mechanics admits a global noncontextual assignment of sharp values. The Kochen–Specker theorem rules out such assignments under its own assumptions [14, 15]. Neither theorem implies that linguistic context should obey a Born rule.

## Appendix D: A minimal quantum attention model

This appendix defines a *minimal quantum attention model*, formulated in a finite-dimensional complex Hilbert space. Its ingredients are learned unitary matrices, normalized complex amplitudes, projective Born readout, and, for the incoherent output, a partial trace. The coherent output retains relative phases and therefore permits phase-sensitive predictions. Together these ingredients define a quantum attention model in its own right, not a reinterpretation of standard transformer attention.

For each target  $i$ , the construction uses an address space  $\mathcal{H}_A^{(i)} \simeq \mathbb{C}^i$  with orthonormal basis  $\{|j\rangle_A : j = 1, \dots, i\}$  and a semantic or value space  $\mathcal{H}_S \simeq \mathbb{C}^V$  with a complete orthonormal vocabulary basis  $\{|y\rangle_S : y = 1, \dots, V\}$ , where  $V$  is the vocabulary size. The complete orthonormal vocabulary basis is an explicit assumption of this toy model. Their joint space is  $\mathcal{H}_A^{(i)} \otimes \mathcal{H}_S$ ; tracing out the address factor leaves a state on  $\mathcal{H}_S$ .

For each  $i$  choose a normalized reference  $|0\rangle_A \in \mathcal{H}_A^{(i)}$  and choose  $|0\rangle_S \in \mathcal{H}_S$ . Encode source position  $j$  and form query, key, and value states by unitary matrices:

$$|\psi_j\rangle = U_{\text{enc}}(x_j)|0\rangle_S, \quad |q_i\rangle = Q_\theta|\psi_i\rangle, \quad |k_j\rangle = K_\theta|\psi_j\rangle, \quad |v_j\rangle = V_\theta|\psi_j\rangle. \quad (D1)$$

Here  $Q_\theta, K_\theta, V_\theta \in U(V)$  are learned. The complex query–key overlap is the score. Independently formed overlaps are not generally one row of a unitary matrix, so they must be normalized explicitly:

$$c_{ij} = \langle q_i | k_j \rangle, \quad Z_i = \sum_{r \leq i} |c_{ir}|^2, \quad (D2)$$

$$\alpha_{ij} = \frac{c_{ij}}{\sqrt{Z_i}}, \quad Z_i > 0.$$

Thus  $\sum_{j \leq i} |\alpha_{ij}|^2 = 1$ . The  $\alpha_{ij}$  are amplitudes rather than classical weights, and their relative phases are retained.

Let one prefix-dependent unitary  $W_i = W_i(x_{\leq i}; \theta)$  prepare

$$|\Omega_i\rangle = W_i|0\rangle_A|0\rangle_S = \sum_{j \leq i} \alpha_{ij}|j\rangle_A|v_j\rangle_S. \quad (D3)$$

The right-hand side is normalized because the address states are orthogonal and every value state is normalized. It can therefore be extended to an orthonormal basis and chosen as one column of a unitary matrix. Operationally,  $W_i$  may be compiled as address-state preparation followed by token-controlled value preparation. In this elementary specification the amplitudes are first obtained from Eq. (D2) and  $W_i$  is then compiled as a prefix-dependent state-preparation unitary; the equations do not provide a coherent overlap-estimation subroutine.

A projective measurement of the address gives the nor-

malized Born attention weights, without a softmax:

$$p_i(j) = \text{tr}[(|j\rangle\langle j|_A \otimes I_S)|\Omega_i\rangle\langle\Omega_i|] = |\alpha_{ij}|^2 = \frac{|c_{ij}|^2}{Z_i}. \quad (\text{D4})$$

If the address is traced out—whether or not it is first measured and forgotten or completely dephased—the retained value state is simply

$$\rho_{S,i}^{\text{mix}} = \text{tr}_A |\Omega_i\rangle\langle\Omega_i| = \sum_{j \leq i} p_i(j) |v_j\rangle\langle v_j|. \quad (\text{D5})$$

Equation (D5) defines the incoherent output of the model. It contains no interference between different addresses. In particular, the partial trace does not produce the coherent sum  $\sum_j \alpha_{ij} |v_j\rangle$ .

From  $|\Omega_i\rangle$ , that sum is obtained by a postselected complementary-basis projective readout rather than by tracing out  $A$ . Put

$$|+\rangle_A = \frac{1}{\sqrt{N_i}} \sum_{j \leq i} |j\rangle_A, \quad N_i = \left\| \sum_{j \leq i} \alpha_{ij} |v_j\rangle \right\|^2. \quad (\text{D6})$$

When  $N_i > 0$ , conditioned on the  $|+\rangle_A$  outcome, the semantic state and the probability of obtaining it are

$$|\phi_i\rangle = \frac{\sum_{j \leq i} \alpha_{ij} |v_j\rangle}{\sqrt{N_i}}, \quad r_i = \text{Pr}(+) = \frac{N_i}{i}. \quad (\text{D7})$$

This postselected projective “quantum eraser” removes which-address information. Measuring which address was selected and coherently erasing that information are alternative readouts of the same prepared state.

The orthonormal vocabulary basis now gives an ordinary projective Born rule:

$$p_i^{\text{coh}}(y | +) = |\langle y | \phi_i \rangle|^2 = \frac{|\sum_{j \leq i} \alpha_{ij} \langle y | v_j \rangle|^2}{N_i}, \quad (\text{D8})$$

$$p_i^{\text{mix}}(y) = \text{tr}(|y\rangle\langle y| \rho_{S,i}^{\text{mix}}) = \sum_{j \leq i} p_i(j) |\langle y | v_j \rangle|^2.$$

Both distributions sum to one. The coherent probability contains phase-sensitive cross-terms; the mixture does not. The parameters of the unitary matrices may be trained by minimizing, for example,  $-\sum_i \log p_i^{\text{coh}}(y_i^{\text{true}} | +)$ . A physical implementation must also account for the

erasure success probability  $r_i$ , for instance by training the joint probability  $r_i p_i^{\text{coh}}(y_i^{\text{true}} | +)$ .

For the promised intervention, retain two orthonormal values  $|F\rangle$  and  $|R\rangle$  for the financial and river-edge senses of “bank.” Let  $\alpha_F = 1/\sqrt{2}$  and  $\alpha_R = e^{i\varphi}/\sqrt{2}$ . Successful address erasure prepares

$$|\phi_\varphi\rangle = \frac{|F\rangle + e^{i\varphi}|R\rangle}{\sqrt{2}}. \quad (\text{D9})$$

For this experiment choose two vocabulary basis vectors as the incompatible pair  $|\pm\rangle = (|F\rangle \pm |R\rangle)/\sqrt{2}$ . The projective Born probabilities are

$$p_+(\varphi) = \cos^2 \frac{\varphi}{2}, \quad p_-(\varphi) = \sin^2 \frac{\varphi}{2}. \quad (\text{D10})$$

Changing  $\varphi$  from zero to  $\pi$  reverses the prediction, while the direct  $|F\rangle, |R\rangle$  populations remain  $1/2$ . By contrast, Eq. (D5) gives the phase-insensitive state  $(|F\rangle\langle F| + |R\rangle\langle R|)/2$  and therefore  $p_+ = p_- = 1/2$ . This is the required experiment in which relative phase changes an observable prediction while direct-path populations are held fixed.

On a classical computer these equations only simulate complex vectors, unitary matrices, a partial trace, and projective Born probabilities. They supply neither physical collapse nor a quantum computational advantage. On quantum hardware  $W_i$ , address erasure, and vocabulary measurement would be physical operations subject to ordinary measurement and no-cloning constraints. The query, key, and value encodings, including Eq. (D3), are freshly prepared from classically specified tokens; they do not copy an unknown quantum state.

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